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	LAB 04(A)
Aim:	Implementation of control structure in R

Theory:	Control statements are expressions used to control the execution and flow of the program based on the conditions provided in the statements. These structures are used to make a decision after assessing the variable. In this article, we'll discuss all the control statements with the examples.
	
	In R programming, there are 8 types of control statements as follows:
	
	if condition
	
	if-else condition
	
	for loop
	
	nested loops
	
	while loop
	
	repeat and break statement
	
	return statement
	
	next statement
	
	Syntax:
	
	❖ if (condition) {
	  # code to be executed if condition is TRUE
	} else {
	  # code to be executed if condition is FALSE
	}
	
	❖ if(condition1){
	# execute only if condition 1 satisfies
	if(condition 2){
	# execute if both condition 1 and 2 satisfy
	}
	}else{
	}

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Output & Program:

sheikh wasimuddin.R

Source on Save

```r
x <- 100
if(x>10){
  print(paste(x,"is grater than 10"))
}
```

5:1 (Top Level)

Console Terminal Background Jobs

R 4.4.0 · ~/

```
R version 4.4.0 (2024-04-24 ucrt) -- "Puppy Cup"
Copyright (C) 2024 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64

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  Natural language support but running in an English locale

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Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

[Workspace loaded from ~/.RData]

> if(x>10){
+   print(paste(x,"is grater than 10"))
+ }
Error: object 'x' not found
> x <- 100
> if(x>10){
+   print(paste(x,"is grater than 10"))
+ }
[1] "100 is grater than 10"
>
```

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sheikh wasimuddin.R*

Source on Save

```r
x <- 5
if(x>10){
  print(paste(x,"is grater than 10"))
}else{
  print(paste(x,"is less than 10"))
}
```

7:1 (Top Level)

Console Terminal Background Jobs

R 4.4.0 · ~/

```
>
>
>
>
>
>
>
>
>
>
>
>
>
>
>
>
>
>
> x <- 5
> if(x>10){
+   print(paste(x,"is grater than 10"))
+ }else{
+   print(paste(x,"is less than 10"))
+ }
[1] "5 is less than 10"
>
```

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sheikh wasimuddin.R

```r
#sheikh wasimuddin 23071044 cseiot 4sem
func <- function(x){
  if(x>0){
    return("is Postitive")
  }else if(x<0){
    return("is Negative")
  }else{
    return("is Zero")
  }
}
func(20)
func(0)
func(-10)
```

10:2 (Top Level)

Console Terminal Background Jobs

R 4.4.0 · ~/

```
>
>
>
>
>
>
>
> x <- 5
> if(x>10){
+   print(paste(x,"is grater than 10"))
+ }else{
+   print(paste(x,"is less than 10"))
+ }
[1] "5 is less than 10"
> func <- function(x){
+   if(x>0){
+     return("is Postitive")
+   }else if(x<0){
+     return("is Negative")
+   }else{
+     return("is Zero")
+   }
+ }
> func(20)
[1] "is Postitive"
> func(0)
[1] "is Zero"
> func(-10)
[1] "is Negative"
>
```

Conclusion:	Hence the Implementation of control structure in R implemented successfully
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